# Reducing Numerical Diffusion in GEMINI3D

Sean Wallace

Department of Physics and Astronomy, Dartmouth College

## Abstract

The choice of numerical solver in plasma simulations, particularly in MHD and hybrid codes, can have a significant impact on the resulting plasma dynamics. However, these choices are often an afterthought in the design of global plasma models, especially in space physics, as researchers tend to focus on data assimilation and visualization. The GEMINI-3D global ionospheric model uses a 5-moment approximation of multi-component fluid transport with an electrostatic constraint. The potential equation is solved with standard finite difference, while the fluid equations are separated into advective and source/diffusion parts, with the advective parts solved by finite volume and the source/diffusion parts solved by the second order TRapeziodal Backward Difference Formula (TR-BDF2). This project attempts to improve the robustness and stability of GEMINI's diffusion solver by replacing TR-BDF2 with Alexander's Two-stage Singly-Diagonal Implicit Runge-Kutta (SDIRK2) method. Specifically, we used a localized front in the $v_z$ drift velocity as a controlled initial perturbation to isolate solver effects from physical phenomena.

## Introduction

### GEMINI3D

- The goal of GEMINI3D is to simulate local ionospheric physics at scales that remain physically interesting (auroral arcs, GDI/KHI, etc). It is used at both middle and high latitudes, with simulation scales of approximately 1000 km$^3$ and $50 \times 50 \times 50$ grid resolution. GEMINI is coupled to atmospheric models to incorporate interactions with neutrals at low altitudes and solar flux models at high altitudes.[5]
- Recent GEMINI development has been primarily focused on two areas: data-assimilation (incorporating sources like PFISR, SWARM. and ASI for data-driven simulations) and improving links to neutral-wind models.[4][2] Numerical methods have been generally neglected since the advent of the model.

### TR-BDF2/SDIRK

- TR-BDF2 is a one step, second order method that combines Crank-Nicholson and backward Euler. Specifically, the update rule is:

$$\frac{f^{n+\gamma} - f^n}{\gamma \Delta t} = \frac{1}{2}(\mathbf{A}^{n+\gamma} f^{n+\gamma} + \mathbf{A}^n f^n) \tag{1}$$

$$\frac{(2-\gamma) f^{n+1} - \gamma^{-1} f^{n+\gamma} + \gamma^{-1}(1-\gamma)^2 f^n}{\Delta t} = (1-\gamma)\mathbf{A}^{n+1} f^{n+1} \tag{2}$$

  This method is technically L-stable, but can contain spurious oscillations (although they are generally damped).[3] [**?**]his method is not singly diagonally implicit, which means the diagonal elements for all stages of the solution are identical.
- SDIRK2 is a *two stage*, second order S-stable method. The two stages and update rule are:

$$\mathbf{Y}_1 = y_n + \Delta t \gamma f(\mathbf{Y}_1) \tag{3}$$

$$\mathbf{Y}_2 = y_n + \Delta t((1-2\gamma) f(\mathbf{Y}_1) + \gamma f(\mathbf{Y}_2)) \tag{4}$$

$$y_{n+1} = y_n + \Delta t \left( \frac{1}{2} f(\mathbf{Y}_1) + \frac{1}{2} f(\mathbf{Y}_2) \right) \tag{5}$$

- SDIRK2 is S-stable when $\gamma = 1 \pm \frac{1}{2}\sqrt{2}$, unlike TR-BDF2. Formally, this means the method is L-stable *and* for any bounded function $g$:

$$\left| \frac{y_{n+1} - g(t_{n+1})}{y_n - g(t_n)} \right| < 1 \tag{6}$$

  L-stability considers only a single test equation, S-stability is about the whole system amplification behavior. Thus for strongly stiff system, SDIRK methods are preferable.[1]

## Method

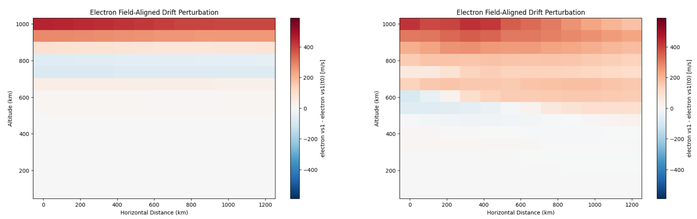
- We modify the PDEparabolic.f90 and diffusion.f90 files in the GEMINI source code to admit choosing SDIRK2 as the diffusion solver. All other methods, including advection solvers, are left untouched. As far as I can tell, this primarily affects the temperature and heat flux equations in the 5-moment approximation.
- We initialize a low-resolution, 2D equilibrium GEMINI run at 67.11°N/212.95°E ranging from 80-1000km in altitude. The grid resolution was $12 \times 1 \times 15$. The simulation time was approximately 16 hours with data writes every 30 minutes. Electrodynamics were ignored. In 2D, this took around 10 minutes to run on 6 MPI processes on my laptop (16GB RAM). Initial conditions were perturbed in the altitudinal electron drift velocity. A Gaussian bump was chosen of the form:

$$\delta v_z^e = 200 \exp \frac{-(z - z_0)^2}{16} \tag{7}$$

  where the denominator was chosen to spread the peak over two grid cells.
- My original plan was to induce an Alfvenic response in the lower ionosphere by inducing a time-varying FAC at the topside of the grid, but this turned out to be very involved and difficult to resolve even at high resolutions. I also played around with neutral temperature perturbations, but had similar luck. The artificial Gaussian bump was the best way to separate solver dynamics from real physical responses.

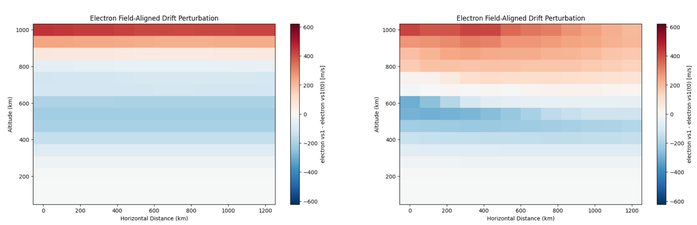
## Results

We observe *substantial* reductions in numerical diffusion and significantly improved preservation of structure at very low resolution with SDIRK2.




((a)) t=73000 ((b)) t=160000

Figure 1: Bump Evolution: TR-BDF2




((a)) t=73000 ((b)) t=160000

Figure 2: Bump Evolution: SDIRK2

## Results

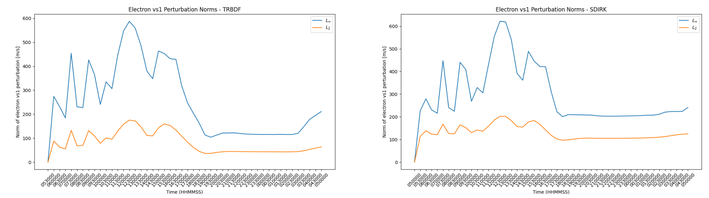



Figure 3: Drift velocity norm over time

- Qualitatively, comparing the norms of the altitudinal drift velocities demonstrates lower numerical diffusion (energy doesn't smear out at late times) and smoother solutions (less jagged peaks/sharp gradients in the middle of the simulation).

## Discussion/Next Steps

- Initial results are promising for improving low-CFL resolution through numerical methods in GEMINI.
- Surprising that small change to numerical method would have such a substantial impact on numerical diffusion - some authors suggest S-stability is not all that important.[3]
- Next experimental steps: improve test setup, look at instabilities (KHI, GDI, FBI). Higher frequency wave modes (approaching Alfven frequency). Look at 3D.
- Next theoretical steps: characteristic analysis, stability analysis. Look at higher order SDIRK methods (although fourth order SDIRK methods are unstable[1]).

## Acknowledgements

- Thanks to Professor Mahlmann for the great course and for the advice on designing this project. Thank you to Charlie for helping me orient myself with GEMINI and to the people who write the genuinely impressive documentation for the model (I think it's mostly Matt Zettergren).
- For transparency - Fortran code for this project was almost entirely generated by GPT 5.4 Thinking. I originally planned to teach myself Fortran as part of this project, but ran out of time.
- Code is available at:

## References

[1] Roger Alexander, *Diagonally Implicit Runge-Kutta Methods for Stiff O.D.E.'s*, SIAM Journal on Numerical Analysis **14** (1977), no. 6, 1006–1021.

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[4] J. van Irsel, K. A. Lynch, A. Mule, and M. D. Zettergren, *Generation of Top-Boundary Conditions for 3D Ionospheric Models Constrained by Auroral Imagery and Plasma Flow Data*, Journal of Geophysical Research: Space Physics **129** (2024), no. 8, e2024JA032722.

[5] M. Zettergren, *Gemini-docs/formulation at main · gemini3d/gemini-docs*, https://github.com/gemini3d/gemini-docs/tree/main/formulation.